The rise of colligations

English *can't stand* and German *nicht ausstehen können*

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This article examines the lexically parallel English and German constructions *can't stand somebody/something* and *jemanden/etwas nicht ausstehen können* "not tolerate (someone or something)", from synchronic, diachronic, and quantitative perspectives. Syntactic and semantic restrictions suggest that the usage of *stand* and *ausstehen* in the relevant sense is older than other semantically similar verbs (e.g. English *tolerate*, German *leiden*), while quantitative evidence from corpora shows that the *can't stand* and *nicht ausstehen können* constructions are both colligationally stronger than lexical competitors. Evidence from the history of *stand* indicates that the lexeme *stand* in the Germanic and other Indo-European languages has a long history of being employed in the relevant sense. The restrictions on usage and the colligational strength of the respective English and German constructions are thus argued to result from the antiquity of the construction and functional competition from other lexemes.

Keywords: collostructional analysis, historical linguistics, Indo-European linguistics, Germanic languages, syntactic productivity

1. Introduction

The following study examines the synchrony and diachrony of the idiomatic phrases English *can't stand somebody/something* alongside the synonymous German *jemanden/etwas nicht ausstehen können* "not be able to out-stand someone/something"; we employ the label "*can't stand* construction" for these constructions together. Examples of this construction in English, German, and the parallel construction in Dutch are given in Table 1; some related usages of *stand*, *ausstehen*, and *uitstaan*, and the similar Swedish *utstå* are shown in Table 2.

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Table 1. Synchronic constraints on E. *can't stand*, G. *nicht ausstehen können*, Du. *niet uitstaan kunnen*

	CAN NOT VERB$_{INF}$ X$_0$ OBJ	*VERB$_{FIN}$ X$_0$ OBJ (P)
English	*She can't stand him.*	**She stands him.* Certain conditions may license *she stands him*, e.g. in a negative polarity context with subordinate clause: *I don't know how she stands him.*
German	*Sie kann ihn nicht ausstehen.*	**Sie steht ihn aus.*
Dutch	*Zij kan hem niet uitstaan.*	**Zij staat hem uit.*

Table 2. Related usages or idioms with the lexical node E. *stand*/ G. *ausstehen*/ Du. *uitstaan* "endure" license non-negated/finite forms

English	*(I can't stand him, but)* she ***can*** *stand him.* [Contrastive focus construction.] See also *to stand out against sth.* ≈ *to withstand sth.*, (OED, s.v. *stand*, sense 59; phrasal verb *stand out*, sub 3)
German	*Ich* ***habe*** *Todesängste* ***ausgestanden****.*
Dutch	*Ik* ***heb*** *doodsangsten* ***uitgestaan****.* "I have gone though mortal agonies. I have been scared to death."
Swed.	*Jag* ***får utstå*** *mycket lidande.* "I have to endure much pain."

Characteristic of this construction in English, German, and Dutch is that the verb *stand*, either with the local particle *out*, as in German and Dutch (and perhaps in older English; see further Section 4), or without it, as in Present-day English, transitively governs a direct object, even though *stand* in its core semantics is an intransitive verb of motion. Furthermore, this idiom exhibits three synchronic constraints in German, English, and Dutch (exemplified in Table 1): first, the idiom's node (E. *stand*, G. *ausstehen*, Dutch *uitstaan*) in the sense of "tolerate", must, under most conditions, be infinitival; second, the infinitival form is tendentially governed by the modal verb *can*; finally, *can* is nearly always negated. More abstractly, the construction in which *stand* serves as a node may be schematically represented (for English) as CAN NOT VERB$_{INF}$ X$_0$ OBJ where *stand* fills the VERB$_{INF}$ slot; X$_0$ stands for other constituents that might intervene between the verb and the (accusative) object. The same semantics of "endure, tolerate" is effectively barred with non-negated, finite *stand*, although non-negated non-finite *stand* may occur under certain further conditions. However, searches in the COCA (Davies, 2008–) and the morphologically tagged corpora of the DeReKo (Kupietz, 2019), on which see further Section 5, revealed no credible instances in English or German of finite, non-negated usages of *stand/ausstehen* with the relevant semantics (outside of the specific collocation *Todesängste ausstehen* in German). On the other hand, instances of non-finite, non-negated *stand* in English subordinate clauses headed by *how*, *if*, or *whether* are relatively well-attested, as well as in subordinate clauses standing under a verb such as *think* or *figure* (e.g. *I figure (that) I can stand it*), or a question with *how*. In German, a handful of instances occur in which *können* does not stand under the negative operator *nicht*, but in which negative polarity is introduced through *kein* "no" or *wenig* "few", e.g. *Umfragen zufolge können immer weniger Briten die Regierungspartei ausstehen* "According to opinion polls, ever fewer Britons are able to tolerate the governing party" (St. Galler Tagblatt, 28.11.2007, p. 6).

This paper aims to provide a diachronically oriented account of why *stand* and *ausstehen* are subject to such restrictions, closely bound up with the CAN NOT VERB$_{INF}$ X$_O$ OBJ construction, whereas other semantically similar verbs, such as E. *tolerate* or G. *leiden* "suffer", are not. However, the constraints as shown in Table 1 are manifestly secondary. They must have arisen diachronically later, because related usages of the same lexical node E. *stand* / G. *ausstehen* / Du. *uitstaan* "endure" still license non-negated, finite forms of the same verb. Moreover, the related Swedish verb *utstå* "endure, suffer" is not generally subject to the restrictions observable in the West Germanic languages.

The syntactic and semantic restrictions found for E. *stand* in the sense of "endure, tolerate" and its cogeners may constitute evidence for a lexical collocation (i.e. preferred lexical co-occurence) between *can not* and *stand*, as well as a colligation between *stand* and the transitive CAN NOT VERB$_{INF}$ X$_O$ OBJ construction mentioned above. On the definition of colligation, see Gries (2009: 14), who defines the term as "the co-occurrence of word forms with grammatical phenomena", such that certain collocations are bound to specific, idiosyncratic grammatical configurations (see also Lehecka, 2015: 5).

This paper will investigate the extent to which the syntactic gaps (evident in both English and German) observed above, as well as morphological gaps (in the case of German *ausstehen*, which virtually exists only as an infinitive or past participle), may be indicative not only of synchronic collocational strength, but also of the relative age of a collexeme. Specifically, in this case, a lexeme that is demonstrably older than semantically similar lexemes takes on more and stronger colligational restrictions.

The organization of the paper is as follows. Section 2 deals with possible synchronic and diachronic factors that cause the emergence of colligations. Colligations may be conditioned synchronically by pragmatic principles or have a diachronic basis. An important diachronic conditioning factor is lexical renewal and competition which can potentially induce older inherited phrases to adopt formal and semantic specializations. In Section 3, further comparison between *can't stand* with other verbal lexemes that can be employed in meanings similar to that of the *can't stand* construction shows that the *can't stand* construction is subject to a number of syntactic and semantic restrictions that do not obtain for those other verbal lexemes. Comparative Indo-European reconstruction shows that features of the lexeme *stand* can be traced back into Proto-Indo-European, where the usages of "stand" in the Germanic languages are compared to functions of cognate lexemes in other Indo-European languages. The etymologies of functionally competing lexemes, meanwhile, reveal them to be, in almost all cases, younger than the core of *stand/ausstehen*. Section 4 then undertakes a lexicographic examination of *stand* and *ausstehen*, from Old English and Old High German into the present day, alongside etymologically and functionally related verbs, documenting the parallel rise of the same colligations in English and German. Finally, in Section 5, we further substantiate the claim that functional competition from semantically similar lexical items has led to the emergence of the robust collocations *can't stand somebody/something* and *jemanden/etwas nicht ausstehen können*, arguing from the perspective of quantitative corpus linguistics that colligations and restrictions may be indicative of relative linguistic age. We demonstrate that the usage restrictions found with *can't stand/nicht ausstehen können* correspond to higher degrees of quantitatively measurable colligational strength. Appendix 6 finds indications that *can't stand/nicht ausstehen können* also exhibit relatively lower degrees of syntactic productivity.

2. Synchronically and diachronically conditioned colligations

How is the emergence of a colligation – like that between G. *ausstehen* and x_O OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN – insofar as it involves paradigmatic or syntagmatic constraints, to be explained? One key to answering this question comes from the insight that colligations like E. *can't stand* and G. *nicht ausstehen können* constitute partially automated units of speech production. Bolinger (1976) and proponents of usage-based approaches (e.g. Bybee, 2010) have pointed out the utility and economy of linguistic formulas or open-slot multi-word constructions, which automate the production of communicatively important and hence frequent semantic concepts, thereby also facilitating language acquisition.[1]

Crucially, the automization of so-called 'chunks' involves 'freezing', i.e. the suspension of a lexeme's free modifiability at the morphological and combin-

1. As Bybee and Scheibman (1999: 577) put it, "[i]n chunking (Haiman 1994), a frequently repeated stretch of speech becomes automated as a processing unit".

ability at syntactic level (see Hackstein, 2012a: 89 for diachronic exemplification). The observation that linguistic automization involves linguistic freezing, and that automization readily applies to frequently repeated stretches of speech, has long been known; this insight is implicit in Meillet's Principle (Meillet, 1937: 31–32: "[r]elics often occur in the most commonly used expressions of a language."), which points to the propensity of linguistically frequent forms to show linguistic anomalies. Freezing need not, however, depend on frequency. Another preserving effect independent of high token frequency is embedding in multi-word expressions, which contribute to the diachronic petrification of grammatical features. The importance of multi-word expressions or "formulas" in general for natural languages was recognized by Firth (1964); see Bolinger (1976), Wray (2002), and Bozzone (2014) for a recent survey of the literature. Moreover, the petrification of older linguistic features in multi-word formulas has revealed itself to be a universal phenomenon in natural languages (see Bybee's research on chunking and the conservation effect; Bybee, 2006, 2007, 2010). The conserving effect of formulaic frames has also been substantiated in other fields of linguistics, for instance in studies on child language acquisition; for example, Arnon and Clark (2011) demonstrated that children acquire irregular morphology first in formulaic contexts. The same applies to poetic languages (*Kunstsprachen*), which notably preserve traces of older morphology and older syntactic constructions in formulaic expressions; see Bozzone (Forthcoming). For a survey of previous literature, see Appendix 1.

Nonetheless, the utility of automization and formularity do not exhaust the conditioning factors that may cause colligations to arise. It is necessary to distinguish between synchronic and diachronic factors, which will be laid out in Sections 2.1 and 2.2.

2.1 Synchronic colligational restrictions

First, colligational restrictions may be described synchronically, the level on which they are part of a speaker's lexical and grammatical knowledge. Consider the constraint on negation observed for E. *can't stand* and G. *nicht ausstehen können* in Table 1. This constraint may be explained by the pragmatic tendency to encode negative emotions indirectly rather than directly. Many languages prefer to encode these speech-acts of disapproving, declining, or rejecting indirectly by negating the positive, thus *I don't approve* or *I can't approve* instead of *I disapprove*, or *I don't like* rather than *I dislike/hate* preferring a syntactically more complex periphrasis. The maxim "Don't address the unpleasant, undesirable directly, and distance yourself from it" holds crosslinguistically and diachronically. For example, senses of antipathy among Homeric Greek heroes or characters in Platonic dialogues are expressed significantly more often by negated οὐκ ἔτλη [oːk étlɛː] "I cannot suffer (it)" than by a non-negated psych verb with a negative denotation. In the same vein, E. *can't bear*, G. *kann nicht aushalten* is pragmatically more apposite than *hate/hassen*. In other cases, limiting verbs to negated expressions might add to vernacular expressivity. Sinclair's example *It doesn't budge* (Sinclair, 1998: 16–22), may fall under this rubric, because *It doesn't budge* conveys more emotional involvement than the nonnegated *It is immobile*.

2.2 The diachronic conditioning of colligations

Beyond the synchronic workings of a given colligation, colligations are frequently conditioned diachronically. Two diachronic mechanisms may be observed that can block normally productive morphological and syntactic derivations.

2.2.1 *Lexical renewal and competition induce linguistic specialization via colligation*

The first mechanism involves gradual lexical renewal without replacement and lexical competition. When an innovative lexeme or phrase impinges upon the meaning or function of a semantically competing existing word or phrase, then the two elements may come into competition with one another. As a result, innovative limitations on the selection of grammatical categories across their constituents on a phrasal level may arise for one of the competing elements, with the consequence that certain morphosyntactic derivations may be blocked.

Lexical renewal typically takes effect gradually, often with a phase in which the inherited lexical item and the innovative lexical item coexist. Thus, the renewal of a lexeme within a multi-word phrase may, and perhaps typically does, undergo lexical renewal without fully replacing the inherited variant of the phrase in question containing the older lexeme in all contexts. In such cases, a principle of economy is operative: semantic differentiation is more economical and useful than synonymy and redundancy. Semantic differentiation entails a semantic repartition, which alots a broader sense to the innovative and usually more frequent expression but narrows the meaning of the same phrase's less-frequent forerunner. This phenomenon constitutes a phrasal analogue to Kuryłowicz's 4th "Law" of Analogy (Kuryłowicz, 1945). Furthermore, the semantic specialization of multi-word phrases is part and parcel of the chunking process, in which semantic specialization and morphological freezing represent each other's analogues on different grammatical levels.

This "Kuryłowicz-4" mechanism of lexical renewal, which prompts chunking and grammatical restriction of the inherited phrase, is arguably exemplified by E. *can't stand* and G. *nicht ausstehen können*. These fall into the semantic domain of psych verbs, which are prone to high degrees of lexical variability and renewal. For instance, alongside E. *can't stand* we find *doesn't like*, *dislikes*, *doesn't suffer*, *can't bear* and alongside G. *nicht ausstehen können*, there is *nicht leiden können*, and *nicht mögen*. The data on synchronic morphosyntactic restrictions imposed upon E. *stand* and G. *ausstehen* treated in Sections 3 and 4 suggest that such restrictions may have arisen due to lexical renewal.

2.2.2 *Diachronically persistent parameters motivate blocking effects and colligations*

The second mechanism giving rise to colligations involves diachronically persistent parameters, which can block productive morphosyntactic derivation. Constructional inheritance can impede otherwise expected morphosyntactic derivations. One may speak in German of *eine abgelegene Hütte* "a remote hut", while the attributive use of German *weit weg* "far away" is excluded: G. **die weit wege Hütte*. Such a ban on the attributive use of G. (*weit*) *weg* is a case of persistence. G. *weg* is diachronically not a true adjective but arose secondarily from a prepositional phrase meaning "on the way"; see (1).

(1) Prepositional phrase "on the way": MHG *ën-wëc* [ɛnˈvɛk], by apheresis > *wëc* [ˈvɛk] > ModG *weg* [ˈvɛk] "away" with prosodic conservation of [ɛ] (contrast ModG noun [ˈveːk]; compare Yid. *avek*, *vek*. See generally *Oxford English Dictionary* (OED; Oxford University Press, 1884–) s.v. *away*, Grimm's *Deutsches Wörterbuch* (DWB; Grimm, 1854–1961) s.v. *weg*).

Certain inherited idioms like E. *far away* and G. *weit weg* typically occur in spoken vernacular registers, which agrees with the empirical fact that one of the primary vehicles both for linguistic innovation and for the transmission of persistent archaisms is language acquisition. This is in accord with Labov (1989: 96), who pointed out that children "reproduce the historically preserved variable patterns", acquiring them as sociolinguistic variables (Labov, 2001: 89). An example of a purely orally transmitted, non-standard variable in English is the alternation between [æsk] and [æks] in some American English dialects, which descends from and replicates the Old English variation between *āscian* and *ācsian* (Labov, 1989: 86).

The same phenomenon is demonstrable for multi-word expressions (collocations, instantiations of particular lexemes in a construction). Examples include (2) and (3).

(2) PWGmc. **an wegan dō-* > E. *to do away (with sth)*, G. colloquial *(etwas) wegtun*, Dutch *(iets) wegdoen*.

(3) PIE **u̯oi̯-/u̯ai̯-* $d^{h}eh_{1}$- > E. *do woe*, OE *wa dydan* (OED, s.v. *woe*), G. colloquial *wehtun*, OHG *tûot wê* (Notker); further Goth. *wai-dedja* "evil-doer".
(see Hackstein, 2012b: 16, Pinault, 2015: 170)

The oral transmission of persistent features over long periods of time also applies to the phrases under discussion, E. *can't stand*, and G. *nicht ausstehen können*, which, as emotive expressions, belong primarily to a vernacular register, and are tendentially replaced by E. *can't bear/tolerate* and G. *nicht leiden/ertragen können* in more formal registers. It can thus be argued that the transitive use of E. *can't stand*, and G. *nicht ausstehen können* is due to inheritance from PIE (see Appendix 2); the otherwise intransitive use of E. *stand* and G. *stehen* is blocked by inheritance.

3. Colligations, restrictions, and the linguistic age of E. *stand* and G. *ausstehen*

The comparison of E. *can't stand* and G. *nicht ausstehen können* with verbs and phrases of parallel formation across the Germanic languages shows that the lexically competing verbs and phrases differ in their linguistic age. The following trend emerges: diachronically older lexemes occur primarily in the context of strong colligations and exhibit more restrictions on their licit co-occurrence, while comparatively younger lexemes exhibit fewer colligations and restrictions, and yet younger lexemes may lack colligations and grammatical restrictions altogether.

The construction E. *can't stand* and G. *nicht ausstehen können* is the oldest; it can, in essence, be traced back to PGmc. (see Tables 1 and 2 above) and partially projected even further into Proto-Indo-European. The second oldest lexical competitor of G. *ausstehen* considered here is *nicht leiden*, which cannot have been used to mean "endure" before the end of the 1st Millennium CE. Finally, the youngest competitor is G. (*nicht*) *mögen* in the sense "(dis)like", which arose in the relevant sense in the New High German era.

In examining a potential correlation between linguistic age and colligational strength, the data in Table 3 shows that the presumably oldest lexeme, *ausstehen*, is highly restricted (five restrictions) and essentially confined to just a single colligation, occurring almost exclusively in collocation with *nicht* and *können*; the second oldest, *leiden*, shows three colligations, and some restrictions on its use (Table 4), while the most recent, *nicht mögen*, exhibits neither any clear colligations nor any apparent grammatical restrictions (Table 5). These facts suggest that the fixity of colligations to some few or just one collocation and the increase in restrictions can be indicative of linguistic age. Although a stable colligation need not be of great antiquity, the establishment of a restricted colligations may be indicative of linguistic archaism relative to lexemes or phrasal units that unrestrictedly enter into productively built phrases. In sum, the stronger the colligation and greater the number of grammatical constraints, the older the construction may be.

Table 3. Colligations and Restrictions for G. *ausstehen*

(PIE >) G. *nicht ausstehen können*: 1 strict colligation involving 5 restrictions		
✓	a. Negated present tense auxiliary construction	Sie kann ihn nicht ausstehen. “She cannot stand him.”
🙁	b. **Nonnegated* present tense auxiliary construction	*Sie kann ihn ganz gut ausstehen.* *“She can stand him quite well.”
🙁	c. *Negated present tense **non-auxiliary** construction	*Sie steht ihn nicht aus. *“She does not stand him.”
🙁	d. ***Nonnegated** present tense **non-auxiliary** construction	*Sie steht ihn aus. *“She stands him.”
🙁	e. ***Nonnegated** past tense **non-auxiliary** construction	*Sie stand ihn aus. *“She stood him.”
🙁	f. ***Nonnegated** perfect passive **non-auxiliary** construction	*Er ist wohl ausgestanden. *“He is stood well.”

* A reviewer holds that German non-negated *ganz gut ausstehen* may be well-formed, though it is the intuition of the native speaker Author 1 that it is not. The phrase “ganz gut ausstehen” does not occur whatsoever in either of the two tagged subcorpora of the DeReKo (Tagged-C and Tagged-C2) employed in Section 5, nor in the considerably larger subcorpus W – Archiv der geschriebenen Sprache, whereas “ganz gut leiden” occurs in those three subcorpora together 73×. Similarly, Google searches (on July 15, 2020) reveal only 582 tokens of “ganz gut ausstehen” versus more than 107000 (!) tokens of “ganz gut leiden”. It seems therefore altogether possible that “(ganz) gut ausstehen” may be sufficiently rare as to be ungrammatical or nearly so for some (perhaps most) native speakers.

While the sort of restrictions found on *ausstehen* as seen above may give some indication as to the relative age of competing linguistic structures, which may be further substantiated through quantitative analysis (see Section 5), the absolute age of a lexeme and importantly, the age of its semantics, can be gauged only by philological facts (chronology of attestation) and the application of comparative reconstruction (this section and Appendix 2). Along these lines, the history and etymology of E. *stand* “endure” and its lexical competitors shows that (i)

Table 4. Colligations and Restrictions for G. *leiden*

(OHG >) G. *nicht leiden können*: 3 weak colligations, 3 restrictions	
√a. Negated present tense auxiliary construction	Sie kann ihn nicht leiden. "She cannot suffer him."
√b. **Nonnegated** present tense auxiliary construction	Sie kann ihn ganz gut leiden. "She can suffer him quite well."
🙁 c. *Negated present tense **non-auxiliary** construction	*Sie leidet ihn nicht. *"She does not suffer him."
🙁 d. ***Nonnegated** present tense **non-auxiliary** construction	*Sie leidet ihn. *"She suffers him."
🙁 e. ***Nonnegated** past tense **non-auxiliary** construction	*Sie litt ihn. *"She suffered him."
√f. **Nonnegated** perfect passive **non-auxiliary** construction	Er ist wohlgelitten. "He is suffered well = he is well-liked."

Table 5. Colligations and Restrictions for G. *mögen*

G. *(nicht) mögen*: no colligations, no restrictions	
√a. Negated present tense auxiliary construction	Sie soll ihn (angeblich) nicht mögen. "She supposedly does not like him."
√b. **Nonnegated** present tense auxiliary construction	Sie soll ihn (angeblich) mögen. "She supposedly likes him."
√c. *Negated present tense **non-auxiliary** construction	Sie mag ihn nicht. "She does not like him."
√d. ***Nonnegated** present tense **non-auxiliary** construction	Sie mag ihn. "She likes him."
√e. ***Nonnegated** past tense **non-auxiliary** construction	Sie mochte ihn. "She liked him."
√f. **Nonnegated** perfect passive **non-auxiliary** construction	Er ist wohl (von den meisten) gemocht worden. "He has probably been liked (by most people)."

E. *stand* "endure" (ii) *like*, and (iii) *endure*, *tolerate* synchronically reflect lexical competitors along a trajectory of descending time depth, and the same holds for

(i) G. *ausstehen*, (ii) *leiden*, (iii) *mögen*. More extensive details on the following are available in Appendix 3.

E. *stand* (*out*) and G. *ausstehen* "*to stand up against, endure sth." are represented in at least three Indo-European branches (Germanic, Slavic, Indo-Iranian; cf. Vedic *úd sthā-* "stand up [against an obstacle]"); Slavic (see Russian *vosstat'* "to rise against, withstand"); and Proto-Germanic **us-standa-* continued in Goth. *us-standan* "rise, be resurrected (with endurance)", and in OHG *ir-standan* "rise, be resurrected", OHG *irstān* "rise; endure", OE *astandan* "stand up; abide; withstand". NHG *ausstehen*, Du. *uitstaan*, E. *stand out* "endure", Sw. *utstå* represent an inherited particle verb construction Gmc. **standa- ūt*, **sta(j)i/a- ūt* which, due to its vernacular register, remained unattested in the OHG and OE literary transmission (Sections 4.2 and 6 below). Furthermore, within West Germanic, the status of the preverb as inseperable and unstressed hints at the compound's linguistic age. Differently, OE *astandan* and ME *ostonden* lose their unstressed preverb (*a-/o-*), merging formally with *stand*, while remaining semantically and collocationally distinct. The anomalous transitivity of E. *stand* "tolerate, endure", G. *ausstehen* continues a persistent parameter which is inherited from the verb's stem formation, see Appendix 2 sub b) and c). Both verbs are barred from their intransitive use, which otherwise is standard for G. *stehen* and E. *stand (upright)* (a blocking effect by persistence, as per Section 2.2.2 above). In contrast to E. *stand* (*out*) and G. *ausstehen*, their lexical competitor G. *erleiden*, *leiden* "to experience, incur sth.", E. *not like* (as stative psych verb) is confined to Germanic and therefore cannot be reconstructed beyond Germanic. Rather G. *leiden* with the sense "tolerate" reflects a semantic innovation (cf. Goth. *ga-leiþan* "come, go") not older than the 8th century CE. Finally, the third and linguistically youngest layer is made up of E. *endure*, *tolerate*, G. *mögen*. E. *endure* and *tolerate* are loanwords, first attested in English in the 14th and 16th centuries, respectively. G. *mögen* in the sense "like" is recent, having arisen in the Modern German period.

4. Colligations in English *can't stand* and German *nicht ausstehen können*

Thus far, we have mustered evidence that E. *can't stand* and G. *nicht ausstehen können* exhibit more colligational effects than do other verbs that permit similar readings in a construction with a negated modal verb *can/können*. These phenomena, as laid out above in Section 3 for G. *ausstehen*, may be argued to correlate with linguistic age; that *can't stand* and *nicht ausstehen können* contain a lexical core that is older than many competing lexemes that may occur in the construction is born out by the reconstructable prehistory and etymology of those lexemes (see Section 3). With these facts in mind, we now turn to the philological investigation of the respective English and German lexemes and constructions, beginning from the Old English and Old High German periods. The objective is to document the functional development of *stand* and *ausstehen*, focusing on emerging colligational restrictions in connection with modal verbs and negation. By and large, the OED and DWB contain sufficient data for the documentation of the relevant colligational restrictions; these are supplemented with further data and lexicographical assessments for English from Bosworth and Toller (2021; hereafter "Bosworth-Toller"), the *Dictionary of Old English* (DOE; Cameron et al., 2018), the Dictionary of Old English Web Corpus (DOEC; dePaolo et al., 2009), and Middle English Dictionary (MED; Lewis, 1952–2001), and, for German, the Referenzkorpus Altdeutsch (RA; Donnhauser et al., 2015), the Referenzkorpus Mittelhochdeutsch (RM; Wegera & Klein, 2016), and the *Mittelhochdeutsches Wörterbuch* (MWB; Hirzel, 2006–). Further details, textual examples, and statistics are provided in Appendix 4.

4.1 *English*: (can't) stand someone/something (out)[2]

The original core semantics as a **verb of position**, meaning "take, have or keep an upright position" (see OED s.v. *stand* I1a) is attested from the Old English period. Example (4) shows that combination with negation and a modal verb, *magan* "be able, can" was entirely possible.

(4) **Ne mæg** hūs nāht lange **standan** on þām heān munte…
(Boethius, *De consolatione philosophiae* 12.26.30; DOEC)
"A house cannot stand for long on that miserable hill…"

A semantic expansion to a **verb denoting physical persistence or endurance** is already found already in Old English as well, as illustrated in Example (5).

(5) … gif satanas (*sic*) winð ongen hine sylfne he bið tōdǣled & he **standan ne mæg** ac hæfð ende (*Gospel of Mark* 3:26; DOEC)
"… if Satan (*sic* SG, not PL) starts up against himself, he is divided, and he cannot stand, but rather has an end."

Of the 244 instances of the form *standan* (usually infinitive) in the DOE, 28 instances are attested as part of a negated modal construction (like *standan ne mæg* in (4)), predominantly with *magan* "can, be able, may", but also *motan* "must; can" and *sculan* "must, should". The constructional components that ulti-

2. All examples in this section stem from the OED (accessed November 4, 2019), unless otherwise indicated, and are cited following the information given in the OED.

mately grow into colligational restrictions are thus already present at an early date. However, clear transitive uses of OE *standan*, or any instances with the sense “endure, tolerate”, are difficult to come by, perhaps entirely absent, in the Old English corpus.

Meanwhile, prefixed OE *a-standan* (where *a-* < PGmc. **uz-* > OHG *ir-*, as in OHG *irstantan*, Goth. *us-* in *usstandan*) is attested with the meaning “stand firm, resist destruction; survive, endure, persevere”; as in Example (6). This verb is not yet, however, clearly susceptible to the reading “tolerate” (see DOE s.v. *astandan*). Likewise, any clear usages of a phrasal verb **standan ūt* “stand out = tolerate” are unknown in the Old English corpus.

(6) seo studu … þe sēo molde on hongode, sēo gesund & ungehrinen from þæm fȳre **astod** & āwunade
(Bede, *Ecclesiastical History* 8.180.29; DOE s.v. *astandan*)
“that post… on which the dust hung in the linen cloth remained safe and untouched by the fire”

Middle English *stō̆nden*, as reflected in the MED, largely continues the usage of OE *standan*, but the sense “make a stand in battle, stand and fight, resist and enemy in battle; withstand” is much more prominently represented, constituting some of the earliest syntactically transitive instances of a form of *stand*. Instances of the infinitive under a negated modal verb or with other negative polarity item are well-represented; examples include (7) and (8).

(7) His strok **may no man stand** (c. 1330, *Sir Tristem*, 2335; MED)
“No man can (with)stand a blow from him.”

(8) There **myghte none** stande hym **a stroke**.
(1470–85 Malory *Morte d’Arthur* x. lxxiv. 543)
“There could no one (with)stand a blow from him”

This transitive usage of *stō̆nden* is entirely parallel to a prominent usage of *astō̆nden* “stand one’s ground, offer resistance; withstand, resist (an enemy, a blow, etc.)”, as per the MED. Given that the *a-* prefix of English (< PGmc. **uz-*) was often aphericized (see OED s.v. *a-*, prefix$_1$), it is possible that some senses of *stand*, especially relating to physical endurance, derive from Old English *astandan*, which has fallen together with *stand* from the Middle English period (the latest citations of *astand* in the OED are c. 1400, in the MED c. 1500). From the Middle English period, *astand* appears transitively in senses approaching “tolerate” (see also MED s.v. *astō̆nden*); an example is shown in (9).

(9) Theih bien londes and ledes, **ne may** hem non **astonde**.
(c. 1330 *Political Songs of England*, 338; MED)

"They are lands and peoples (which) no one can tolerate."

Transitive usages of *stand* in the EEBO corpus (1470–1700; Davies, 2017) between 1600 and 1700 predominantly attest to resistance or endurance of an explicitly physical kind, e.g. *stand a blow*, *stand a fight* (see Appendix 4 for a fuller list). Such instances are possible with or without negation, and with or without a modal verb; see Example (10) below.

(10) volp: o god, sir! i were a wise man, **would stand the fury of a distracted cuckold** (1616, *The workes of Beniamin Ionson*; EEBO)

From c. 1600, transparent usages of *stand* as a psych verb, including with pronominal objects (equivalent to the Present-day English usage shown in Tables 1 and 2) are first attested, and a semantically equivalent expression *stand out* OBJECT first appears.[3] Interestingly, *stand out* OBJECT (~ 120× in the EEBO) tends to fill the object slot with noun phrases belonging to the very same semantic sphere – an adverse thing or event – as the objects with *stand* OBJECT above; see Examples (11) and (12). Corpus searches for instances of *stand out* (ARTICLE) NOUN/PRONOUN and *stand* (ARTICLE) NOUN/PRONOUN *out* in the COCA (Davies, 2008–), COHA (Davies, 2010–), BNC (Davies, 2004), and Hansard (Davies, 2015), strongly suggest that *stand out* ADVERSE EVENT/THING was seemingly moribund by the 19th century (at least in American English and the British Parliament): it occurs but only 4× or 5× in COHA (4× 19th century, perhaps only one 20th century instance) and 1× (1888) in the Hansard corpus. The latest unambiguous attestations of *stand out* (ARTICLE) NOUN/PRONOUN known to us are found in 1949 (COHA), where *stand out the night* occurs twice (thanks to an anonymous reviewer for bringing this attestation to our attention).

(11) i **can not stand** nosing of Candlestickes, or euphuing of similes, alla sauoica. (1593, *Pierces supererogation*; EEBO)

(12) if noble lisbon **could not stand it out**, where is that city so resolv'd, and strong.... (1655, *The Luisad, or Portugals historicall poem*; EEBO)

The data from Early Modern English indicate that the reading of *stand* as a psych verb, indicating **mental/emotional endurance**, had emerged by at least this period, and the usage of *stand (out)* in the relevant sense became increasingly restricted to the context provided by the CAN NOT VERB$_{INF}$ X$_{O}$ OBJ construction in the 18th and 19th centuries, as evidenced in Examples (13) and (14).

3. Note that this usage must be distinct from *stand out* in the sense "be noticeable; be better", e.g. *the bright lettering stands out well from/against a dark background*; *her work stands out from the rest as easily the best*, which remains in use in Present-day English.

(13) It is often said, such an **one cannot stand** the Mention of such a Circumstance.
(1710 R. Steele *Tatler* No. 225. 2)

(14) Soapy **couldn't stand out** against the big ranchmen when they got together and meant business. He had to pull his freight.
(1936, W. Raine *Crooked Trails and Straight*; COHA)

Finally, note that Present-day (British) English preserves *stand out* "endure" in the idiom *stand out against something*, which can be paraphrased as "continue to resist", as in (15).[4]

(15) I have had **to** stand out **with** my editor once or twice on that point.
(1887 G. R. Sims *Mary Jane's Mem*. 296)

4.2 German: From *(ir)stan(tan)* to *nicht ausstehen können*[5]

As in English, the cognate OHG verbs *stān* and *stantan* are well-attested as basic **verbs of position**, and in contexts that admit of readings relating to **physical endurance**, seen in (16).

(16) Fóra sinen óugon **stent** alle ménnisgon, úbile joh gúate.
(Otfrid, *Evangelienbuch* 5.20; RA)
"Before his eyes stand all people, evil and good."

As mentioned under 3, the particle verb *ausstehen* (MHG *uȝstân*) is first directly attested in the Middle High German period, but only with senses other than "endure, tolerate", as the example in (17) shows.

(17) dô muost mein herr etlich tag dô still ligen, das die pferd ein wênig **ausstunden**. (c. 1470 *Tetzel Rozmit*.160; MWB)
"you have to remain for a few days, my lord, so that the horses might rest (*ausstunden*) a little"

Nevertheless, the OHG prefixed verb *irstantan*, whose primary sense is "stand up, arise", is susceptible to the reading "endure" in some instances, although employed exclusively intransitively, as, for instance, in (18).

(18) Soso uúas Ionas in thes uuales uuámbu thrí taga inti thriio naht... Thie Nineuiscun mán **arstantent** in tuome mit thesemo cunne inti furniderent iz...
(Tatian 57; RA)

4. Compare further the fixed English idiom *stand the test of time*, as well as Modern Icelandic *standast próf* "to pass an exam".

5. Citations in this section stem from the DWB (accessed through http://woerterbuchnetz.de/DWB (4 November 2019), unless otherwise indicated.

> "So Jonah was in the belly of the whale for three days and three nights... The man from Nineveh endured in the trial with this creature and humiliated it..."

Similarly, MHG *erstân* knows at least one clear instance in which this verb is clearly to be read in the sense "endure, tolerate", shown in (19).

(19) ir habt gestriten einen strît, / daz mich immer wunder hât, / wie ez iwer lîp **erstât** (*Garel von dem blüenden Tal* 8325–7; Walz, 1892)
"you have fought such a battle, that it amazes me, how your bodies stand it"

From the late 16th century, *ausstehen* comes to be attested in abundance (the earliest occurrence in DT: 1583), at which point its primary sense is "endure, tolerate"; an early such usage is given in (20). It may be inferred that *ausstehen* perhaps belonged primarily to the vernacular register of the language, being clearly documented once its primary sense became that of a psych verb, and sources became more abundant. *ausstehen* has conceivably replaced a possible function of older OHG *irstantan* / MHG *erstân* in the sense of "endure".

(20) das er von dem verbotenen Bawm gessen, so hilfft es jn doch nicht sondern mus die gedrawte straff **ausstehen**
(1583, L. Polycarp, *Ein Christliche Leichpredigt* 23; Berlin-Brandenburgischen Akademie der Wissenschaften, 2020)
"that he ate from the forbidden tree did him no good; rather, his wife had to endure punishment"

Older German does preserve traces of the inherited use of *ausstehen* as intransitive "step out", as seen under (21) below, and has developed the transitive phrasal verb *etwas ausstehen* "to out-stay, endure something or somebody", as in (22). Like English *to stand (out)*, *ausstehen* has adopted three restrictions: to the negated form (Restriction 1, seen in (23)), use with a superordinate modal verb (Restriction 2, seen in (24)), and the restriction of the modal verb to *can* (Restriction 3, illustrated in (25)).

(21) So schied er ab von Babylon, in die insel Sagena kam, da stund man aus in gottes nam. (16th c., H. Sachs I, 171d; see also DWB 1.985, s.v. *ausstehen* sub 1)
"So he departed from Babylon, and came to the island of Sagena, where one appeared (= "stood out") in god's name"

(22) nach allen prüfungen, die ihr **ausgestanden** habt
(18/19th c., Goethe 14, 196; see also DWB 1.985, s.v. *ausstehen* sub 4)
"... after all of the trials that you have endured"

Restriction 1: negative polarity item, colligation with negation, giving *steht nicht aus*:

(23) er **steht** den schmerz **nicht aus**, er überwältigt ihn. (18th c., Zachariä 1, 179)
"he cannot tolerate the pain, it overwhelms him"

Restriction 2: restriction to infinitival use after modal verbs *kann, darf … nicht ausstehen*:

(24) das kein unglück so grosz ist, es sei geistlich oder leiblich, das ich **nicht künde ausstehen** und überwinden. (Luther 6, 345a)
"… that no misfortune, whether spiritual or physical, is so great that I cannot endure it and overcome it"

Restriction 3: further restriction of modal verb to *kann*, *kann nicht ausstehen*:

(25) sie konnte ihren herrn vater **nicht** eher **ausstehn**, bis er u. s. w. (18/19th c., Klinger 1, 168)
"she could not stand her father, until he etc. …"

4.3 Summary of attested diachronic developments

The available evidence points towards the diachronic development of restrictions on the English (phrasal) verb *stand* (*out*) "bear, tolerate" and German *ausstehen* along the following path, binding themselves to transitive negated modal verb constructions:

i. **Semantic narrowing:** *stand (out)* and *ausstehen* are originally verbs of position, used to denote physical endurance, but which have undergone a semantic narrowing to become psych verbs denoting a negative psychological attitude towards a person, thing, or experience. *stand (out)* and *ausstehen* have likely replaced older forms OE *astandan* and OHG *irstantan*, containing a prefix etymologically related to E. *out*, G. *aus*.
ii. **Negative polarity:** Both verbs adopt a constraint of occurring only in the negated form; the non-negated use persists only in fixed idioms like *Ängste (Todesängste) ausstehen*.
iii. **Negated modal verb and nonfinite constraint:** both *stand* (*out*) "bear, tolerate" and *ausstehen* are increasingly restricted to infinitives under the negated modal *can/können*.

5. Quantitative analysis of colligations, cohesion, and linguistic age

Considering the deeper historical background of *stand (out)* and *ausstehen* in the foregoing, further evidence pointing towards a strong colligational relationship between the CAN NOT VERB$_{\text{INF}}$ X$_{\text{O}}$ OBJ construction and the verbs *stand/ausstehen* in Present-day English and German can be mustered. In this section, synchronic evidence that speaks in favor of clear and well-established colligations will be systematically assessed; this evidence of colligational strength is in turn argued to be indicative of the colligation's relatively greater antiquity.

5.1 Collostructional analysis and constructional productivity

Potentially, cohesion between the lexemes *stand* or *ausstehen* and the relevant English and German constructions could become evident prosodically through the univerbation or phonetic reduction of elements in the phrase, independent of any regular synchronic phonological processes or broader sound changes in progress. Since, however, no transparent phonetic or phonological traces of chunking are available for the phrases *can't stand* and *nicht ausstehen können*, evidence pointing towards a strong colligational relationship between the relevant lexemes and the construction must be assessed from a purely quantitative perspective, with the help of large-scale corpora. The methods applied here fall under the label of COLLOSTRUCTIONAL ANALYSIS (Gries & Stefanowisch, 2003; Evert, 2004: Chapter 3).

To compare the colligational behavior of the phrases under discussion with synonyms or semantically similar lexemes is useful to this end. For instance, the collostructional strength of English *can't stand sb./sth.* can be compared with that of *can't tolerate someone/something*. For German, the collostructional strength of *jdn./etw. nicht ausstehen können* can be compared with that of *jdn./etw. nicht leiden können*, etc. These are specific instantiations of the more general, abstract construction characterized by the presence of a transitive verb in its infinitival form standing under the modal verb CAN and negation, which was introduced in Section 1: CAN NOT VERB$_{\text{INF}}$ X$_{0}$ OBJECT$_{\text{ACC}}$.

The synonymous usages of this construction that concern us here (*can't stand someone/something*, *can't bear someone/something*, etc.) can in turn be compared with a sample of other non-synonymous verbal collexemes (e.g. *make*, *kill* and their German counterparts) in the construction, to provide a fuller picture of the extent to which verbs such as *stand* and *bear* cohere with this construction. Section 5.2 presents the data, quantative collostructional analysis, and assessment for for English (5.2.1) and German (5.2.2), respectively. Details on the procedures of data collection are available in Appendix 5. Likewise provided in Appendix 6 is an examination of the PRODUCTIVITY of specific instantiations of the *can't stand* construction (e.g. CAN NOT STAND$_{\text{INF}}$ X$_{0}$ OBJECT, CAN NOT TOLERATE$_{\text{INF}}$ X$_{0}$ OBJECT, etc.), following the methodology for the corpus-based measurement of morphological productivity developed in Baayen (1989, 1992) and further applied to larger syntactic units in Zeldes (2012) and Bozzone (Forthcoming). That investigation is intended to test the hypothesis that constructions with quantitively lower relative productivity may be suspected of being diachronically older.

5.2 Colligational strength in the CAN NOT VERB$_{INF}$ X$_0$ OBJECT$_{ACC}$ construction: Data collection and quantitative analysis

The question to be quantitatively investigated here is: how strongly does a lexeme like *stand* or *tolerate* prefer to occur in the context of the CAN NOT VERB$_{INF}$ X$_0$ OBJ construction? Intuitively put: colligational strength is greater the more that the tokens of a lexeme occur in a given construction, beyond what the rates of occurrence of the lexeme and the construction would independently predict.

How best to determine the degree of collocational (in the case of word sequences) or colligational (in the case of partially abstract syntactic constructions) strength remains a debated issue. Conservatively, one might conclude that no single statistic decisively represents collostructional/colligational strength in absolute terms. For this reason, several different statistics will be reported and interpreted. Examination of the correlation between various association measures in Levshina (2015: 235–239) indicates that there may be three major categories: measures that indicate the dependence of a collexeme on a given construction (e.g. RELIANCE of Schmid, 2000, or ΔP with collexeme cue of Ellis, 2006), measures that indicate how strongly a collexeme is drawn to a construction (e.g. ATTRACTION of Schmid, 2000, or ΔP with constructional cue of Ellis, 2006), and some measures that indicate general cohesion between a collexeme and a construction (e.g. Pearson's Chi-Squared Test or the chi-squared log-likelihood ratio test). In this connection, the general conclusion of Wiechmann (2008) is worth reporting: his MINIMUM SENSITIVITY POINT ESTIMATE (the smaller of RELIANCE and ATTRACTION) showed the best correlation to a regression model fit to a psycholinguistic experiment concerned with identifying fixed processing units (i.e. idioms or constructions; see Kennison, 2001, for these results). See further Evert (2004: especially Chapter 4) on the mathematical properties of assorted association measures.

To present a full portrait of collostructional strength, we report in this section all association measures mentioned above. For calculating these statistics, four values are needed: the frequency of the lexeme in the construction, the frequency of the lexeme overall, the frequency of the construction overall, and the frequency of constructions containing the POS (part of speech) of the lexeme (i.e. the frequency of all verbal constructions). These quantities reach into the thousands or hundreds of thousands of occurrences in the corpora consulted, and therefore certainly contain some false positives and false negatives.

5.2.1 *English*

Quantitative data on the token frequency of the relevant construction, with specific verbal collexemes, the token frequency of those verbal collexemes, and the token frequency of verbal constructions in general in English was harvested from the Corpus of Contemporary American English (COCA; Davies, 2008–).[6] Table 6 reports the total token frequency of the specific verbal collexemes in the transitive CAN NOT VERB$_{INF}$ construction and the corresponding token frequency of the verbal lemma in the COCA.[7] 2×2 contingency tables for each verbal collexeme were then constructed, to allow for the calculation of association measures. The contingency table for *stand* (Table 7) is given as an example.[8]

Table 6. Frequency of English Verbal Lemmata and Occurrence in the CAN NOT VERB$_{INF}$ X$_{O}$ OBJECT$_{ACC}$ Construction

Verbal collexeme	CAN NOT VERB$_{INF}$ X$_{O}$ OBJECT$_{ACC}$ Freq.	Verbal lemma Freq.
Stand	2266	197108
Tolerate	340	7866
Bear	818	29943
Do	7341	3676671
Make	5093	1219896
Kill	337	134186
Love	147	159950

On the basis of the respective contingency tables, the following summary statistics as measure of colligational strength were calculated for each verbal collexeme: RELIANCE and ATTRACTION (Schmid, 2000), ΔP with both collexemic and constructional cue (Ellis, 2006), MINIMUM SENSITIVITY POINT ESTIMATE (Wiechmann, 2008), Pearson's chi-squared statistic, and the chi-squared log-likelihood ratio (*G*-Test). All statistics were calculated using *R* version 3.5.2 (R

6. All queries were first conducted on May 1, 2019, then later updated and corrected on October 17 and October 21, 2019.

7. In the case of *stand* and *bear*, manual checking of the data allowed us to exclude tokens that did not correspond to the semantic reading "cannot endure, cannot tolerate", e.g. the collocations *bear witness* and *bear children* in the case of *bear*.

8. A reviewer observes that the value obtained for the token frequency verbal constructions (100535787) in the *COCA* may not be entirely accurate, since the query used (see Appendix 5) likely results in an overcount. Since this same value was used in constructing the contingency tables for all collexemes, the general tendency of the statistics, and the qualitative conclusions derived therefrom, should not be adversely affected.

Table 7. Contingency Table for *stand* and the CAN NOT VERB$_{\text{INF}}$ X$_{\text{O}}$ OBJECT$_{\text{ACC}}$ Construction

	CAN NOT Cx	¬ CAN NOT Cx	TOTAL
Stand	2266	194842	197108
¬ *Stand*	158908	100179771	100338679
TOTAL	**161174**	**100374613**	**100535787**

Core Team, 2018); Pearson's chi-squared statistic and the likelihood ratio were obtained using the function `assocstats()` from the package *vcd* (Meyer et al., 2017); the other measures of colligational strength were calculated using functions written by the second author. All statistics are reported rounded to five decimal places, reported in Table 8.

Table 8. Measures of Colligational Strength for English Verbs in the CAN NOT VERB$_{\text{INF}}$ X$_{\text{O}}$ OBJECT$_{\text{ACC}}$ Construction

Verbal collexeme	**ATTRACTION**	**RELIANCE**	**ΔP collexeme**	**ΔP construction**	**MINIMUM SENSITIVITY**	**Pearson's χ^2**	**χ^2 likehood ratio**
stand	0.01406	0.0115	0.00991	0.01212	0.0115	12076.5	5071.4
tolerate	0.00211	0.04322	0.04162	0.0020	0.00211	8514	1599.9
bear	0.00508	0.02732	0.02572	0.00479	0.00508	12374.7	3122.7
do	0.04555	0.002	0.00041	0.00899	0.002	369.17	343.28
make	0.0316	0.00417	0.0026	0.0195	0.00417	5102.9	3545.1
kill	0.00209	0.00251	0.00091	0.00076	0.00209	69.256	58.993
love	0.00091	0.00092	−0.00069	−0.00068	0.00091	46.844	55.416

The surveyed collexemes may be divided into four groups on the basis of measures of colligational strength: *stand* alone, *tolerate* and *bear*, *do* and *make*, and *kill* and *love*. The last group, *kill* and *love*, is only very weakly associated with the CAN NOT VERB$_{\text{INF}}$ X$_{\text{O}}$ OBJECT construction, and *love* is in fact repelled from it (see ΔP). For all collexemes surveyed here other than *love*, more tokens were observed in the construction than would be independently expected. *kill* and *love* thus exhibit patterns of usage largely independent of the construction. *Make* and *do*, meanwhile, exhibit the highest degrees of attraction to the construction, but the comparatively lower degrees of reliance. Conversely, *bear* and *tolerate* show the highest degrees of reliance, but comparatively lower degrees of attraction. Finally, in the case of *stand*, this collexeme is neither as reliant on the CAN NOT VERB$_{\text{INF}}$ X$_{\text{O}}$ OBJECT construction as its synonyms *bear* and *tolerate*, nor as strongly attracted to the construction as the very high frequency lexemes *do* and *make*. Precisely because *stand* shows neither a very small value for ATTRACTION nor a small value for RELIANCE, it has the highest MINIMUM SENSITIVITY value of all collexemes surveyed (exceeding 0.01).

These patterns largely replay themselves in the χ^2-statistics, where *stand*, *tolerate*, *bear* exhibit the largest values Pearson's χ^2-test, while *do* and *make* have comparatively larger likelihood ratio values, and *kill* and *love* the smallest values. *Bear*, in fact, shows the largest χ^2-value overall (12374.7), but which is not considerably larger in the range of values seen here than the value for *stand* (12076.5). The χ^2 likelihood-ratio test points in a similar direction, where *kill* and *love*, followed by *do* return the smallest values; *make*, however, obtains a larger value than that of *tolerate* and the value for *stand* is largest (5071.4).

Overall, these results lend themselves most readily to the interpretation that *stand* is most strongly associated with the CAN NOT VERB$_{\text{INF}}$ X$_{0}$ OBJECT construction among all of the collexemes surveyed. In particular, *stand* appears to outrank its potential synonyms in this construction, *tolerate* and *bear*, in terms of ATTRACTION and the degree to which the construction cues the collexeme *stand*. On the other hand, because *tolerate* and *bear* are considerably less frequent than *stand* (see Table 7) above, their usage is more dependent upon the construction under investigation. With respect to the control collexemes *do*, *kill*, and *love*, *stand* and *bear*, and to a lesser extent, also *tolerate* and *make*, are clealy more strongly associated with CAN NOT VERB$_{\text{INF}}$ X$_{0}$ OBJECT$_{\text{ACC}}$.

From these quantitative results, we can draw the conclusion that *stand* in the sense typically assumed by *can't stand sb./sth.* has a stronger connection to the CAN NOT VERB$_{\text{INF}}$ X$_{0}$ OBJECT than most verbs of English. This comparatively strong colligational association is then to be attributed to the restrictions placed on *stand* in the relevant sense discussed in Sections 1 and 3. If a phrasal verb **stand out* directly cognate to German *ausstehen* and Dutch *uitstaan* were in use in contemporary English, the English results for **stand out* might then more closely resemble the quantitative picture seen for German *ausstehen* in 5.2.2 below, with greater dependence upon the CAN NOT VERB$_{\text{INF}}$ X$_{0}$ OBJECT$_{\text{ACC}}$ construction – the reason that *stand* does not exhibit greater dependency on this construction is partly a consequence of the generally high frequency of this lexeme, which occurs in a wide range of distinct usages.

5.2.2 *German*

To acquire frequency data on the colligation under investigation for German, queries were applied to morphologically tagged subcorpora TAGGED-C and TAGGED-C2 of the Deutsches Referenzkorpus (DeReKo; Kupietz, 2019) through the COSMAS-II interface.[9] In contradistinction to English, the fact that, in German, different constituent orders occur in main clauses as opposed to (most types of) subordinate clauses, different query terms were needed to accommodate and find these different possible word orders. In the following, we represent the relevant construction for German as X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN, since we take the constituent order in subordinate clauses to reflect the underlying order of constituents prior to the V-to-T-to-C raising (V2; see Holmberg, 2015) of tensed verbs and verbal auxiliaries in main clauses. At least one further option is syntactically possible in main clauses: the topicalization or focusing of the verbal infinitive in initial position of a main clause, giving rise to structures such as *Ausstehen konnte er sie überhaupt nicht* "stand her, he absolutely could not". This structure in an abstract form proved difficult to query properly in COSMAS-II, and searches with concrete verbal collexemes revealed it to be very uncommon. Therefore, instances of X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN with topicalization/focus of the infinitive are excluded from the results below.

Table 9 reports the total token frequency of the specific verbal collexemes in the transitive X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN construction and the corresponding token frequency of the verbal lemma in the TAGGED-C and TAGGED-C2 subcorpora of the DeReKo.

Table 9. Frequency of German Verbal Lemmata and Occurrence in the X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN Construction

Verbal collexeme	X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN Freq.	Verbal lemma Freq.
ausstehen "stand (out)"	718	5938
aushalten "tolerate"	150	13775
leiden "suffer"	672	88363
ertragen "bear, endure"	397	16231
tun "do"	809	674154
machen "make"	2728	2651446
töten "kill"	41	98558
lieben "love"	95	75608

2×2 contingency tables for each respective verbal collexeme were then constructed. The contingency table for *ausstehen* is given as an example (Table 10). Just as with the English data above, summary statistics as measures of colligational

9. All queries were first conducted on May 8, 2019, then updated and corrected on October 20, 2019.

strength were then calculated, using the same array of statistics explained. Results are reported in Table 11.

Table 10. Contingency Table for *ausstehen* and the X_O OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN Construction

	NICHT VERB$_{INF}$ KÖNNEN Cx	¬ NICHT VERB$_{INF}$ KÖNNEN Cx	TOTAL
ausstehen	718	5220	5938
¬ *ausstehen*	360524	260791754	261152278
TOTAL	**361242**	**260796974**	**261158216**

Table 11. Measures of Colligational Strength for German Verbs in the X_O OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN Construction

Verbal collexeme	ATTRACTION	RELIANCE	ΔP collexeme	ΔP construction	MINIMUM SENSITIVITY	Pearson's χ^2	χ^2 likehood ratio
ausstehen	0.00199	0.12092	0.11954	0.00197	0.00199	61423.1	5090.3
aushalten	0.00042	0.01089	0.00951	0.00036	0.00042	901.20	358.41
leiden	0.00186	0.0076	0.00622	0.00152	0.00186	2477.1	1195.4
ertragen	0.0011	0.02446	0.02308	0.00104	0.0011	6257.6	1540.9
tun	0.00224	0.001	−0.00018	−0.00034	0.001	16.424	17.198
machen	0.00755	0.00103	−0.00036	−0.0026	0.00103	243.50	267.18
töten	0.00011	0.00042	−0.00097	− 0.00026	0.00011	66.776	92.252
lieben	0.00026	0.00126	−0.00013	0.00005	0.00026	0.87962	0.90774

In contrast to the English data, the German data may first be divided into two groups on the basis of the ΔP statistics: one the one side stand *ausstehen*, *aushalten*, *leiden*, and *ertragen*, which appear to be genuinely attracted to and reliant upon the X_O OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN construction; on the other stand *tun*, *machen*, *töten*, and *lieben*, which are repelled from and occur independently of the construction under investigation. In the case of *tun*, *machen*, and *töten*, the statistically significant χ^2-values result from observed frequencies that are markedly lower than their corresponding expected values (e.g. for *machen*, Oberserved = 2728, Expected = 3668). *tun*, *machen*, *töten*, and *lieben* thus do not build colligations with the NICHT VERB$_{INF}$ KÖNNEN construction.

Conversely, the first four verbal collexemes to appear to build part of colligations with this construction, with varying degrees of strength. The two χ^2-statistics and MINIMUM SENSITIVITY all point to the same overall ranking of colligational strength: *ausstehen* (strongest), *ertragen*, *leiden*, *aushalten* (weakest). In terms of Schmid (2000)'s ATTRACTION and RELIANCE, remarkable is the comparatively high degree of RELIANCE exhibited by *ausstehen* (about 5× greater than its nearest competitor, *ertragen*), and the relatively strong degree of ATTRACTION exhibited by *leiden*. In the former case, high RELIANCE but middling ATTRACTION points to a collexeme that is highly dependent upon the NICHT VERB$_{INF}$ KÖNNEN construction; in diachronic terms, it appears that *ausstehen* has become frozen in this colligation – the verbal lexeme cannot be employed in German with great freedom (see examples in Sections 1 and 3). *leiden*, meanwhile, represents a high-frequency verb that occurs in the context of the the NICHT VERB$_{INF}$ KÖNNEN construction significantly more often than would be expected, hence its relatively greater values for ATTRACTION and ΔP Construction. *ertragen*, then, appears to fall into a place similar to *ausstehen*, although it has not yet become as dependent on NICHT VERB$_{INF}$ KÖNNEN as the latter verb. In summary, the connection between *ausstehen* and the NICHT VERB$_{INF}$ KÖNNEN construction is remarkably strong: *ausstehen* exceeds the value for every other collexeme on every measure of colligational strength employed.

The overall conclusion may again be drawn that X$_O$ OBJECT$_{ACC}$ NICHT *ausstehen* KÖNNEN represents a strong colligation in comparison to other transitive verbs generally in German and potentially synonymous verbs as well. In contrast with English *stand*, however, *ausstehen*'s locus of occurrence is much more dependent upon the relevant construction, whereas English *stand*, while attracted to CAN NOT VERB$_{INF}$ X$_O$ OBJECT$_{ACC}$, is not nearly so dependent on it. As mentioned at the end of Section 5.2.1, the divergent behavior of X$_O$ OBJECT$_{ACC}$ NICHT *ausstehen* KÖNNEN versus CAN NOT *stand* X$_O$ OBJECT$_{ACC}$ must be due to the fact that English *stand* in the relevant sense is morphologically identical to other forms of *stand*, in a wide range of senses. If a phrasal verb **stand out* were still in use in contemporary English, we might predict for it to similarly be very dependent upon the CAN NOT VERB$_{INF}$ X$_O$ OBJECT construction.

6. Conclusion

The present case study on E. *stand* "endure, tolerate" and G. *ausstehen* (as well as Du. *uitstaan* and Swed. *utstå*), as most clearly transmitted in the *can't stand* construction, showcases the lifecycle of a verb and its construction over several millennia. Departing from an original verb of position in PIE, the verb in question comes to denote physical/cognitive endurance, finally becoming a psych verb of (negated) mental/emotional endurance in West Germanic, usually embedded in a specific constructional context. The high degree of colligational strength between E. *stand* in the relevant sense and G. *ausstehen*, respectively, and the can not verb$_{INF}$ x$_{O}$ object$_{ACC}$ construction may be attributed to lexical competition with other innovative verbs of the same or similar meaning, following the "Kuryłowicz-4" mechanism discussed in 2.2.1. Indeed, *stand* "endure, tolerate" and *ausstehen* are effectively limited to occurring in the can not verb$_{INF}$ x$_{O}$ object$_{ACC}$ construction. As shown in Section 3 and Appendix 2, these Germanic lexemes directly continue the PIE verbal root lexeme **steh$_2$-* "stand (upright) firmly", which built a causative stem PGmc. **standa-* active "to cause sb/sth to stand upright", middle "to cause oneself to stand upright (against sb/sth)", and which metaphorically meant "to be able to sustain, to bear, endure". Comparative evidence from Indo-Iranian and Slavic, and across the diachrony of the Germanic languages, suggests that the verb optionally permitted the local and directional preverbs Gmc. **ūt* "on high", and **ut-s* "upward", as orally transmitted, vernacular variables (see Section 2.2.2, Section 4 on E. *stand* (*out*), and G. *ausstehen* as vernacular lexemes, and Appendix 2). PIE **ut-s* "upward" prefixed to **standa-* produced PWGmc. **urstanda-*, the ancestor of OE *astandan* and OHG *irstantan*, which are attested with the sense "endure"; the prefixed verb **urstanda-* likely competed with a vernacular particle verb construction **standa- ūt*, **sta(j)i/a- ūt* which underlies EModE *stand out* and G. *ausstehen*.

Evidence from the OE and OHG corpora indicates that the continuants of PGmc. **standa-* could readily enter into constructions with negation and modal verbs. In West Germanic, the verb with particle **ūt* adopted three colligational restrictions, namely, (i) negative polarity and negated use, (ii) use with modal verbs, involving negation transfer to the modal verb, (iii) restriction of the negated modal verb to E. *can not*, G. *nicht können*, Du. *niet kunnen*. The rise of these three restrictions can be attributed to lexical competition with other etymologically younger lexemes of parallel function, thereby leading to the virtual restriction of *stand* "endure, tolerate" and *ausstehen* to usages in the can not verb$_{INF}$ x$_{O}$ object$_{ACC}$ construction. The contrastive collostructional analysis in Section 5 and the analysis of constructional productivity in Appendix 6 ultimately indicate that there might exist a relationship between the age of a lexeme and the extent to which it becomes reliant upon a particular construction. Naturally, not all lexemes with a long prehistory are destined to become fossilized within a given constructional frame – whether such constructional reliance with concomitant restrictions arises depends precisely upon the emergence of lexical competition. Expanding upon the observation of Kuryłowicz (1945) concerning competing morphological forms, it is then the oldest lexeme among a set of competitors that is expected to be pushed into a specialized function. Whether such a tendency holds more generally requires the investigation of the diachrony of more lexemes and colligations across a wider typological array of languages. In the case at hand, though, the marriage of comparative reconstruction and quantitative corpus linguistics indicates that *stand* in the sense "endure, tolerate" has likely undergone precisely such a development.

The following abbreviations are employed in this article

E.	English
G.	German
Du.	Dutch
OE	Old English
ME	Middle English
(E)ModE	(Early) Modern English
OHG	Old High German
MHG	Middle High German
ModG.	Modern German
Goth.	Gothic
PIE	Proto-Indo-European
(P)(W)Gmc.	(Proto)-(West-)Germanic
(Hom.) Gk.	(Homeric) Greek
Lat.	Latin
(Ved.) Skt.	(Vedic) Sanskrit
Yid.	Yiddish

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Appendices

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Online Appendices for “The rise of colligations: English *can’t stand* and German *nicht ausstehen können*

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Appendix 1. Previous research on the diachronic evolution of synchronic collocations and colligations

Building upon Firth (1964) and Manning and Schütze (1999: 151–189), Gries and Stefanowitsch (2003) have refined the study of collocations through the application of quantitative corpus linguistic methods, including collostructional analysis. Collostructional analysis permits the quantification of the extent to which the usage of a particular lexeme is bound up with the usage of a particular construction. Thus, one can precisely assess whether a lexeme depends upon or triggers the employ of a given construction, which potentially allows for the discovery of diachronic archaisms through the synchronic quantitative assessment of collostructional dependencies. Nonetheless, many insights following from quantitative corpus linguistics have not yet been integrated in the synchronic and diachronic analysis of multi-word phrases; for instance, recent treatments of German phraseology in Palm (1995), Burger (2007a, 2007b), and Donalies (2009) do not attempt to apply any quantitative approaches in their analyses.

Appendix 2. Comparative reconstruction of E. *stand* "tolerate, endure", G. *ausstehen* "bear, tolerate" and persistent parameters

Inherited and persistent parameters of E. *stand* "tolerate, endure", G. *ausstehen* "bear, tolerate" can be identified in three domains: a.) lexical semantics, b) the verb's transitivization through preverbation, and c) its stem formation.

a.) The lexical semantics of PIE **steh₂-* underlying E. *stand*, G. *ausstehen*

The lexical semantics of E. *stand* "tolerate, endure", G. *ausstehen* "bear, tolerate", as well as the verb's polysemy between a verb of position and a verb of physical or cognitive endurance (by metaphorical extension) is found in Germanic, Slavic, Latin, Greek, and Vedic Sanskrit (see LIV^2 s.v. **steh₂-*), which warrants the projection of this polysemy with intransitive "endure" back into the Proto-Indo-European phase.

E. *can't stand sb./sth.* contains the simplex verb *stand* in the metaphorical sense of "be strong and endure," which continues the semantics of the PIE root **steh₂-* "stand (upright) firmly". A similar metaphorical extension, producing the sense "endure" is also found in other, earlier attested Indo-European languages, for instance:

i) Latin **verb of position** *stare* "to stand, be in a standing position, stand up (of people, animals)" (Glare's, 2012, *Oxford Latin Dictionary*, henceforth "OLD", s.v. *stō*, 1), "to stand up (in or for combat)" (OLD, s.v. *stō*, 1 d 2) → **verb of endurance** "to remain stable, last (of a city institution etc.)" (OLD, s.v. *stō*, 16), "to continue to exist, endure" (OLD, s.v. *stō*, 17).

ii) Greek **verb of position** ἵστημι [hístɛːmi] causative "make to stand, set up" (Liddell et al., 1925–1940, henceforth "LSJ", 841, s. v. ἵστημι, sub A causal), anticausative ἵσταμαι [hístamai] (with aorists ἔστην [éstɛːn], ἕστηκα [héstɛːka]) "to be set up, erected" (LSJ, 841, s. v. ἵστημι, sub B II 4) → **verb of endurance** ἵσταμαι [hístamai] "to stand still [and endure]" (LSJ, 841, s. v. ἵστημι B II).

b.) Transitivization of *$steh_2$- through optional preverbation and reinforcement with PIE *ud/ūd/ut-s*

Possibly already in Proto-Indo-European, and certainly in Proto-Germanic, *$steh_2$- could be optionally reinforced with a local preverb: PIE **ud* (Ved. *úd* "up, upon"), **ūd* "on high" (with monosyllabic lengthening in PGmc. **ūt* "out"), and directional *ut-s* "upward" (in OCS *vŭz*, Goth. *us*, ModG. *er-*) can be found in together with *$steh_2$-, thereby yielding a collocation **ud/ūd/ut-s* $steh_2$- "stand (up to sb./sth.) firmly, endure". Examples of the local and directional particles with *$steh_2$- follow below.

i) **Local PIE *ud (Ved. *úd*), *ūd** "on high": Vedic *úd sthā-* "stand up [against an obstacle]", hence "endure, withstand": RV 4.4.4a: *úd agne tiṣṭha* "stand up, Agni [against your enemies]", RV 10.53.8b: *út tiṣṭhata* "stand up (endure)!", RV 8.76.10: *úd tíṣṭhan ójasā́ sahá* "standing up [enduring] with your might". This Vedic usage of *ud sthā-* entirely parallels the sense of "endure" attested throughout the modern Germanic languages.

ii) **Directional PIE *ut-s > *uss > *us.** Examples of directional **ut-s* include OCS *vŭs-stati* (Kurz's, 1958–, *Lexicon linguae palaeoslovenicae*; "LLP", 1.339 s.v. *vŭs-stati* 2) *na kogo* "to rise up against sb." [sich gegen jem. erheben]), Russ. *vosstat'* "to rise against, withstand", Goth. *us-standan* "rise, be resurrected (with endurance)", and OHG *ir-standan* "rise, be resurrected" (cf. further 3.2.3.1 below). The function of the morpheme *-s* in PIE **ut-s* "upward" is not certain (cf. Schneider 2011: 189f., Dunkel 2014: 823f., 825, Opfermann 2016: 250–2, Kroonen 2010: 370), but it is likely directional (cf. Hackstein 2002: 109 fn. 12), or less likely ablatival *"from up(right)"; it is worth noting that PIE *$(h_1)en$-s* means "inwards, into", not "from inside".

Repetitive reinforcement caused PGmc. **uz* to become **ūt-uz-*, which is continued by Gothic *ut-us-*, and OHG *ir-stantan* → *ūz-ir-stantan* (Schützeichel, 2012: 310); cf. ModG *er-koren* "choose", *aus-er-koren* "chosen", ModG. *etwas erstehen* with semantic shift "to reach sth. by endurance". The directional function of **ut-s* as continued in G. *er-*, against the local function of **ūt* as continued in G. *aus-*, also accounts for the fact that the bound prefix G. *er-* occurs as a telic-transitivizing morpheme in German and is more strongly transitivizing than G. *aus-*. Contrast a) G. itr. *steigen* "climb" vs. tr. *er-steigen* "to reach/accomplish sth. by climbing" with b) G. intr. *steigen* "climb" vs. intr. *aus-steigen* "to get out, get off".

Given the intransitive meaning of the base lexeme PIE **steh*$_2$-, Gmc. **stō*- "stand (upright) firmly", the transitive use of E. *stand*, Du. *uitstaan*, G. *ausstehen* as "endure sb./sth." is anomalous and hints at a persistent archaism in accordance with Meillet's (1937) heuristic principle that persistent linguistic structures tend to appear synchronically as anomalies. In each case, however, the proof must be provided by comparative reconstruction. The synchronically anomalous transitivization of **steh*$_2$- in E. *stand*, Du. *uitstaan*, G. *ausstehen* may be attributed to two factors: addition of preverbs **ud* or **ut-s*, and the specific stem formation built in Proto-Germanic (see c.) below).

Table A1 illustrates some instances in both Modern and Old Indo-European languages in which preverbation results in transitivization. Although **ud* and **ut-s* may not have always had a strongly transitivizing effect (cf. Schneider, 2011: 185, and Kulikov, 2012: 734 sub 25 on Vedic Sanskrit *úd*), the West Germanic languages have developed a common optionally transitivizing function: such examples include German *aussitzen* "sit out (smth.)" (< intrans. *sitzen* "sit"), *ausspielen* "finish playing; play somebody off against somebody" (< intrans. *spielen* "play"), or English *outrank* (< intrans. *rank*).

Table A1. Transitivizing effect of preverbs on "go"

Language	**V_{itr}**	**→ V_{tr}**	**Gloss**
German	*gehen*	*über-gehen*	"transgress, ignore"
Russian	*idti*	*pere-jti*	"cross over"
Latin	*gredi*	*ag-gredi*	"go towards, attack"
Hittite	*pai-*	*istarna arha pai-*	"go through" (Cotticelli-Kurras, 2007: 13)

c.) Morphological transitivization of **steh*$_2$- by its stem formation (nasal infix)

Given that the Present-day English transitive usage of *stand* does not clearly continue a virtual **stand out* "endure", another mechanism other than preverbation might be responsible for the transitivization of **steh*$_2$- in Germanic as well. The Proto-Germanic stem formation **standa*-, a nasal-infix present with dental enlargement (see Prokosch, 1939: 67, 157; Lühr, 2000: 94–95; Rix et al.'s, 2001, *Lexikon der indogermanischen Verben*, henceforth LIV^2 591 fn. 1; Mottausch, 2015: 99, and Ringe, 2017: 277) is a likely culprit. PGmc. **standa*-, as a nasal present, continues a PIE causative function "to place", which, in middle voice reads as "to place oneself, step, stand". On the transitivizing function of the nasal present in PIE, see Meiser (1993), Villanueva-Svensson

(2011), Covini (2016/7: 290–328 [326–8]), Zasada (2020). Scheungraber's (2014: 69–75) alternative proposal to derive the stem of Germanic _stand_ from an _nt_-participle is less persuasive because it leaves the verb's lexical semantics unexplained; see also Mottausch (2015: 97).

In the active voice, PGmc. *_standa_- and *_ūt/ŭs standa_- meant "to cause sb/sth to stand upright; to put up, erect", as well as metaphorically "to be able to sustain, to bear, endure". For similar metaphorical extensions using other roots, compare PIE *_tel_h_2- "to lift", "to be able to lift and endure", underlying Hom. Gk. οὐκ ἔτλη [oːk étlɛː] "he couldn't bear", or PIE *b^h_er_- "carry, bear", underlying Latin _non ferō_ "I can't stand (something)". In the middle voice, PGmc. *_stand_- and *_ūt/ŭs stand_- meant "to cause oneself to stand upright (towards)", "to stand up (to)", and metaphorically "to endure"; in addition, the phrase began to be construed with an accusative object denoting the goal, with the meaning "to stand (up) to somebody or something". The goal adjunct came to be reanalyzed subsequently as a direct object, increasing the valency of *_ste_h_2- to that of a two-place predicate. For a Germanic parallel for the transition of a motion verb governing the accusative of goal and its semantic development to a psych verb, see Appendix 3 below on G. _leiden._

Old Norse still semantically mirrors the middle forms in _standa_ "stand, be steadfast", _standask_, _stózk_ "to let oneself stand out against; withstand, endure"; likewise, Modern Icelandic _standast próf_ means "to pass an exam", and _standa sig_ "to maintain one's hold". The same mechanism, namely, the shift from a motion verb to an experiencer verb, also accounts for the transitivity of English _under-stand_, OE _understandan_, Frisian _understonda_, Middle Dutch _understēn_/_understān_ from PIE *(h_1)_n̥tér ste_h_2-, PGmc. middle *_under standa_- "cause onself to stand in between/amidst something, and be able to perceive and comprehend it". Note here that _under_ reflects not PIE *_n̥_d^h_ér_ "under, beneath", but *(h_1)_n̥tér_ "in between", which merged with PIE *_n̥_d^h_ér_ in Germanic; see Harm (2003: 115).

Appendix 3. E. *stand* / G. *ausstehen* and their lexical competitors: history and etymology

The sketch in Appendix 2 of the linguistic prehistory behind E. *stand* and G. *ausstehen* strongly suggests that the functions of these verbs found in the present-day collocations *can't stand* and *nicht ausstehen können* reach back into at least into Proto-West Germanic, but likely even into Proto-Indo-European. A closer look at the lexical competitors of *stand* and *ausstehen* in the sense "endure", E. *tolerate*, *endure*, G.*leiden, ertragen, aushalten, mögen*, whose colligational relationships to the CAN NOT VERB$_{INF}$ X_0 OBJ construction are explored quantitatively in Section 5 of the main text, largely reveals the opposite to be true: their usages as transitive verbs meaning "endure" are comparatively innovative, or they constitute recent lexical borrowings; see the following etymologies.

a) English

<u>stand</u>

Intransitive *stand* is, as shown under Appendix 2, reconstructable for Proto-Indo-European, and well-attested from Old English onward (see Bosworth-Toller, 1921, s.v. *standan*), though transitive usages are first clearly found in the 14th century (cf. MED s.v. *stō̆nden*). Note that English originally possessed a verb *astandan* (see Bosworth-Toller s.v. *astandan*, MED s.v. *astō̆nden*, OED s.v. *astand*), cognate with OHG *irstandan* (cf. above), which may have fallen together formally with *standan* via apheresis of the prefix *a-* (see OED s.v. *a*-, prefix$_1$). Given OHG *irstantan* and Gothic *usstandan*, a Proto-Germanic **uz-standa-* likely existed, and perhaps could have been employed in the meaning "endure, withstand" besides the more robustly attested sense "arise, stand up".

<u>bear</u>

Alongside E. *stand* "endure, tolerate" we find synonymous E. *bear* "endure, tolerate" which, as a lexeme is of equal linguistic age and can be projected back into Proto-Indo-European; cf. Lat. *ferre* "endure, tolerate", Greek φέρω "endure, tolerate" (LSJ 1923 s.v. III, e.g. Homer *Odyssey* 18.135), and Vedic Sanskrit *bhar*- "endure, tolerate" (Grassmann, 1872 [1976]: 956 No. 14 "zu erfahren haben").

In accordance with its linguistic age, E. *bear* "endure, tolerate" exhibits many of the same syntagmatic constraints as E. *stand* "endure, tolerate", see, e.g. the colligation of E. *bear* "endure, tolerate" with *can't*. One cannot say **She bore to be neglected/ being neglected*, while *She can't bear to be neglected/ being neglected* is grammatically acceptable (cf. Bolinger 1976:9). The quantitative collostructional analysis under Section 5 likewise indicates that *bear*, like *stand*, is more attracted to and dependent upon the CAN NOT VERB$_{INF}$ X$_0$ OBJ construction than the other verbs investigated. These two verbs have, however, specialized somewhat differently, in that *stand* exhibits a predilection for pronominal objects, while *bear* prefers infinitival complements.

<u>tolerate, endure</u>

E. *endure*, *tolerate* are loan verbs (see OED s.v. *endure* and *tolerate*), first attested in English in the late 14th and 16th centuries respectively. They thus represent the linguistically youngest layer in English, in contrast to the older, inherited verbs *stand* and *bear*. Consequently, they are largely free of derivational constraints.

b) German

<u>ausstehen</u>

G. *ausstehen* with stressed separable prefix *aus-* has an Indo-European pedigree. It reflects the collocation of the local prefix PIE **ud*, PGmc. **ūt* and PIE **steh₂-*. While in both OHG and MHG, *stān*/*stân* occurs with a separable prefix *ūz*/*uȝ* in a wide variety of meanings (from "jump up" to "rest"), the meaning "endure, tolerate" does not happen to be one of them. By contrast MHG *erstân* (< OHG *irstān*) is attested with the sense "endure, tolerate" (cf. 4.2 in the main text). Therefore, G. *ausstehen* likely replaced older MHG *erstân* (< OHG *irstān*) with unstressed inseparable prefix OHG *ir-* from PGmc. **us* and PIE **ut-s*. The collocation of PIE **ut-s* with PIE **steh₂-* is attested in Germanic, Slavic, and Vedic (cf. Appendix 2).

<u>leiden</u>

G. *leiden* in *nicht leiden können* derives etymologically from the motion verb OHG *līdan*, Goth. *ga-leiþan* "come, go", PGmc. **lei̯þe/a-* (PIE **lei̯th₂-* "go"; see LIV^2: 410). The underlying verb of motion could take an accusative of goal in the sense of "to go to/reach sb./sth.", and metaphorically as "to incur something", with a semantic shift from a motion verb governing the accusative of goal to an experiencer and thus psych verb; cf. G. *leiden* in the idioms *Schaden leiden* “incur and suffer damage”, and *jemanden nicht leiden können* “not be able to tolerate sb.”. The semantics and the construction of G. *leiden* “endure” first arose in the High German of the 8th or 9th century. The verb is not as old as the forebears of G. *ausstehen.*

ertragen

NHG *ertragen* appears to be a comparatively recent coinage. It is unattested in OHG and MHG, and its first attestations stem from the early NHG period. See the DWB 3.1031, s.v. *ertragen*, sub 2 *sustinere, tolerare*.

aushalten

Like NHG *ertragen*, the first attestations of NHG *aushalten* in the sense of “withstand, tolerate” come from early NHG. See the DWB 1.879f., s.v. *aushalten*, *sustinere*, sub 4) mit acc. der person, and sub 5) mit acc. der sache: *den kampf, krieg, angrif, sturm, das spiel aushalten*.

mögen

The same applies to German *mögen* “like” and *nicht mögen* “dislike”, which belong to the semantically most recent layer of the modal *mögen*, arising first in NHG. The senses “like” and “dislike” presuppose the semantic shift of *mögen* “be able”, as attested for OHG *magan*, MHD *mügen* and preserved in NHG *vermögen* “be able”.

Appendix 4. The attested history of English *can't stand* and German *nicht ausstehen können*

This appendix accompanies Section 4 of the main paper and serves primarily to provide further examples of contextual usage of the verbal lexemes under discussion, as well as some basic statistics on frequencies of occurrence in different periods of English and German.

A4.1 English: *(can't) stand someone/something (out)*[4]

Old English *standan* as a **verb of position** is also illustrated in (A1); compare (4) in the main text.

(A1) … þā **stōdan** him twegen weras bīg (*sic*) on hwītum hræglum. (*Ascension Day Homily*, 99–100; DOEC)

"… there stood before him two humble men in white clothing"

(A2) also attests to the semantics of **physical persistence or endurance**; cf. (5) in the main text.

(A2) … and fela samod tugon, ac heō næs āstyrod, ac **stōd** swā swā munt. (*Ælfric's Lives of Saints*, 100; DOEC)

"… and many are drawing together, but she is not moved, but rather stood her ground (lit. mountain)."

(A3) represents another instance of OE *astandan* with the reading "endure"; cf. (5) in the main text.

(A3) forðæm is ðæm lǣce swīðe geornlīce to giemanne ðæt he swā strangne lǣcedom selle ðæm sēocan, swā he mæge ða mettrymnesse mid geflieman, & eft swā līðne swa se tȳdra līchoma mæge **astandan** (*Gregory's Pastoral Care* 61.455.28; DOE s.v. *astandan*)

"Because it is for that doctor very desirable to take care that he give such a strong medicine to the sick one, so that he can drive that illness away with it, and thus can the frail body pleasantly so endure"

Nevertheless, the lexical possibilities for the extended sense of mental or emotional endurance "tolerate, endure" in the Old English are mainly represented by a range of other verbs, including *(a-)beran* "bear" *a-drēogan* "perform, carry out; endure, suffer", and *ar-æfnan* "suffer, endure, hold out". That lexemes with the meaning "endure, tolerate" so often exhibit a prefix related to the preposition "out" suggests that the semantics of "out" could often compositionally construct the semantics of mental endurance. In particular, *aræfnan* seems to have been the default lexeme for expressing the meaning "endure, tolerate" in the Old English period, also forming part of an expression *aberan and aræfnan* "to bear and endure" (cf. DOE s.v. *aræfnan*); this lexeme is, however, evidently extinct by the Middle English period.

Examples (A4) and (A5) complement examples (7) and (8) in the main text, illustrating ME *stō̆nden*.

(A4) Fyrumbras was aggreued sare þat O[lyuer] hym **stod** so longe (c. 1380, *English Charlemagne Romances I: Sir Ferumbras*, 634; MED)

"Ferumbras was sorely aggrieved that Oliver (with)stood him so long"

(A5) Hij ne mowen þat assaut stonde, And fleiȝeþ a litel by þe stronde (c. 1400, *King Alexander* 3732; MED)

"he could not (with)stand that assault, and escaped a little by the beach"

Although the sense "withstand, resist, endure" is found for both *stō̆nden* and *astō̆nden* in Middle English, this usage is unattested in the EEBO corpus (1470–1700) of late Middle and Early Modern English until c. 1600. Transitive usages of *stand* in the EEBO between 1600 and 1700 predominantly attest to resistance or endurance of an explicitly physical kind (spellings here are normalized): *stand a blow*, *stand a/the fight*, *stand the test*, *stand a/the shock*, *stand the charge*, *stood the (enemy's) armies*, *stood the assaults*, *stand a/the trial*, *standing the brunt (of the battle / of continual ceremony)*, *stand the force, stand the fury*. See also Jespersen (1927:345 §16.7,6). Example (A6) complements (9) in the main text.

(A6) …deborah in person out-braving danger, and **standing the brunt of the battell**, against many thousands, living armed and awake… (1640, *The exemplary lives and memorable acts of nine the most worthy women in the world*; EEBO).

While the construction *stand out* OBJECT, appearing after c. 1600, is largely equivalent to *stand* OBJECT, the two differ somewhat in that is that the former is very frequently attested in the form *stand it out* "endure (something)" (2× in the 1580s, 315× (!) in the 17th century), while pronominal objects with simple *stand* are comparatively rare (see two clear instances in (A8) and (A9)) in this period. Examples (A7)–(A11) attest to the same pattern seen in examples (11) and (12) of the main text.

(A7) Hundreds he sent to Hell, and **none durst [dare]** stand **him** (a1616 *Shakespeare Henry VI, Pt. 1* (1623) i. i. 123).

(A8) haru: oh ware what loue is? ned hath found the scent; and if the connie chaunce to misse her Burrough, Shee's Shee's ouer-borne y fayth, she **can not stand it** (1616, *English-men for my money: or, a pleasant comedy called, A woman will haue her will*; EEBO).

(A9) pal: any thing: sink, ruin, perish: fate has not that frown, nor heaven and all its thunder has that bolt, but i **could stand 'em** all for dear jacintha (1697, *The world in the moon, an opera*; EEBO)

(A10) i from those eyes for ever will remove, i **can not stand the sight** of hopeless love (1673, *The Empress of Morocco, a tragedy*; EEBO)

(A11) Jesus fled from the persecution; as he **did not stand it out**, so he **did not stand out against it** (1649 Bp. J. Taylor *Great Exemplar* I. vi. 105).

Given that the *stand* OBJECT and *stand out* OBJECT constructions in Early Modern English still exhibit a predilection for objects that allow the reading "withstand", it seems plausible that these usages directly or effectively continue OE *astandan*. That *stand out* OBJECT should have this same particular preference, even though, as noted above, no OE **(a)standan ūt* or ME **(a)stŏ̄nden out(e)* seems to be attested, is peculiar, unless one presumes that such a particle verb did in fact exist prior to Early Modern English as a vernacular alternative to the *astandan* of the written language.

Alternatively, the formal merger of OE *standan* and *astandan* in *stand* provided an opening for the creation of the particle verb *stand out* as a potential replacement for *astandan*.

Once *stand* had developed the sense of a psych verb indicating **mental/emotional endurance**, the relevant sense became increasingly restricted to the context provided by the CAN NOT VERB$_{INF}$ X$_0$ OBJ construction in the 18th and 19th centuries. For example, the COHA attests 242 instances of CAN NOT *stand* PRONOUN in the period 1800–1900, versus 201 instances of CAN *stand* PRONOUN without negation and certainly yet fewer instances of *stand* PRONOUN, without either negation or a preceding form of CAN. Examples (A12)-(A15) parallel examples (13) and (14) of the main text.

(A12) Till I am satisfied in these particulars, you and I must by no means meet; **I could not possibly** stand **it** (1750 Ld. Chesterfield *Let*. 19 Nov. (1932) (modernized text) IV. 1621).

(A13) **She could not** stand that Manager fellow. **I could not** stand him myself (1879 M. Oliphant *Within Precincts* II. Xx. 60).

(A14) Captain Blunt jumped up. "My mother **can't** stand tobacco smoke" (1919 J. Conrad *Arrow of Gold* iv. Ii. 162).

(A15) Houses… many times **cannot** well **stand out** a long Lease (1676, H. Phillippes *Purchasers Pattern* 18)

For further details on the relevant usage of *stand*, see also OED s. v. *stand*- **V.** Transitive senses, **52.**, **58.**, and **59**. Compare (A16) and (A17) with (14) in the main text on the idiom *stand out* in British English; cf. also Swedish *Jag tar inte ut längre med henne* "I can't stand it with her any longer".

(A16) We managed to stand out against all attempts to close the company down (Hornby & Cowie, 1989:1250).

(A17) He is in good earnest, and will execute these threatnings upon them if **they will** obstinately stand it out with him (*a*1694 J. Tillotson *Serm.* (1742) III. xxxv. 17).

A4.2 German: From *(ir)stan(tan)* to *nicht ausstehen können*[8]

(A18) shows OHG and *stantan* in its sense as a basic verb of position.

(A18) Andares tages abur **stuont** Iohannes inti fon sinen iungiron zuene inti giscouuota then heilant gangantan… (Tatian 16; RA)

“But on the second day John and his two disciples stood and saw the savior going by…”

To example (22) in the main text, it may be added that Pennsylvania German, as an isolated dialect of German, still maintains unrestricted usages of the verb *ausstehen*, i.e., Pennsylvania German *ausschteh*; see Beam (2004–2011) 1 A-152 for the examples in (A19) and (A20):

(A19) Sie will’s geduldich ausschteh.
“She wants to suffer through it patiently.”

(A20) Der Hi hot gemehnt, des waer schlimm, en Mensch so ausschteh losse.
“He was of the opinion that it would be wrong to permit a human being to suffer so.”

Paralleling the restriction of *austehen* to use after modal verbs alongside (24) is (A21), while (A22) pairs with (25), indicating the further restriction to a negated construction with *können* specifically.

(A21) es ist aber einer, der dir deinen trotz wol **kann ausstehen** (16th c., Luther 6, 226b).
“…he is, however, one who can tolerate your contrariness well”

(A22) nein gnädger herr, sie **kann** ihn **nicht ausstehn**. (18/19th c., Tieck 3, 203)
“no, gracious lord, she cannot stand him.”

Appendix 5. Data collection for assessing colligational strength in the CAN NOT VERB$_{INF}$ X$_0$ OBJECT$_{ACC}$ construction

This appendix describes the query procedures used for obtaining the data for colligational analysis in Section 5.2 of the main text from the *Corpus of Contemporary American English* (COCA) and the subcorpora TAGGED-C and TAGGED-C2 of the *Deutsches Referenzkorpus* (DeReKo).

5.1 *English*

To obtain a relatively complete picture of the frequency of different types of elements in the X$_0$ OBJECT slot(s) of the transitive CAN NOT VERB construction, the following parts of speech were searched for in the COCA interface following the VERB slot for each collexeme (the relevant COCA query sequence is given in parentheses): pronoun (_p*), noun (_nn*), adjective followed by a noun (_j* _nn*), possessive pronoun followed by noun, with or without an intervening adjective (_app* [_j*] _nn*), verbal gerund (_v?g*), determiner or quantifier followed by a noun with or without an intervening adjective (_d* [_j*] _nn*), article with or without an intervening adjective (_at* [_j*] _nn*), or negation (_xx*). These possible X$_0$ OBJECT sequences were preceded by ca|can|could _xx* VERBAL-COLLEXEME (e.g. STAND, BEAR) in the search query. Thus, as an example, the complete query ca|can|could _xx* STAND _nn* returns all instances of the sequence *ca*, *can*, or *could*, followed by either *not* or *n't*, followed by any form of the lexeme *stand*, followed by any noun. Queries were run through both the List and Collocates functions of the COCA interface. The total frequencies of the respective verbal collexemes were obtained through queries of the form STAND_v, BEAR_v, etc. An approximate token frequency for all verbal constructions in the COCA was obtained through the query _v*; this search yielded a total token frequency of 100535787. Similarly, an approximate token frequency for the transitive CAN NOT VERB construction was obtained through queries with simply _v* in the slot of the verbal collexeme; this yielded a token frequency of 161174.

5.2 *German*

Since the German verbal collexemes examined here all obligatorily require a direct object, including particular types of nominal phrases was not necessary in the search term; throughout, two different search patterns were employed to find instances of our colligation in main and subordinate clauses, respectively (VERB here is to be read as a verbal lemma, e.g. *ausstehen* or *machen*):

- Main Clause:((&können %w0 (, oder ? oder .)) /+w5 "nicht") /+w1 &VERB /w0 MORPH(V INF)
- Subordinate Clause: "nicht" /+w1 (&VERB /w0 MORPH(V INF) %w0 (, oder ? oder .)) /+w1 &können

For a main clause, the search term finds all instances of the lemma *können* followed by *nicht* and a given verbal lemma in its infinitive, with up to five words separating *können* and *nicht*; punctuation (an orthographic indication of a clause boundary) cannot intervene between *können* and *nicht*. For a subordinate clause, the search term finds all instances of *nicht* immediately followed by a given verbal lemma in its infinitive, in turn immediately followed by a form of *können*; punctuation cannot intervene between the infinitive and *können*. To obtain a count of all instances of verbal forms in the two subcorpora, the simple search MORPH(V) was applied, yielding 261158216 tokens. Searches for the X$_0$ OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN construction generally returned 362916 tokens, where the search term MORPH(V INF)was used without the specification of a lexeme.

Appendix 6: Productivity, colligational strength, and the age of constructions

It is a recognized fact that higher degrees of cohesion and greater fixity of an idiom (i.e. fewer possibilities to modify the elements of a multi-word construction) are indicative of comparatively older constructions, all things being (semantically and pragmatically) equal; cf. the well-researched correlation between chunking and morphological freezing (Bybee, 2006, 2007, 2010) and univerbation (Hackstein, 2012a), and the notion of *Festigkeit* "cohesion, fixity" in idioms according to Burger (2007a, 2007b). For similar results on Homeric Greek, see Bozzone (Forthcoming). Hence, a construction with a quantitively lower relative productivity may be reasonably regarded as diachronically older. This does not mean that "old" constructions are necessarily "unproductive" in the sense that they cannot generate new types altogether, but merely that they may be expected to be comparatively *less* productive than other constructions that fulfill the same function. In the case at hand, we indeed find that higher colligational strength (on a variety of measures) correlates with lower productivity (as measured by Baayen's $\mathcal{P}$; see Baayen, 1989, 1992). This correlation suggests that particular collexemes ride similar diachronic waves, in both becoming more bound to particular constructions and exhausting the potential for the construction to generate new types.

The basic notion, as applied by Bozzone (Forthcoming), is that diachronically older constructions (and in particular, constructions that are "in decline" and are in the process of being replaced by functionally similar constructions), tend to lose, rather than gain, types, while a few established instances of the construction represent an increasing proportion of the tokens. This view then lends itself to the application of the corpus-based measures of morphological productivity developed by Baayen (1989, 1992; see also Sandell, 2015, in the context of historical linguistics), in particular the statistic $\mathcal{P}$ (= HAPAX LEGOMENA / TOKENS), as a means of potentially comparing the relative age of functionally parallel, competing constructions.

Zeldes (2012) has already previously demonstrated that the same corpus-based measures of morphological productivity developed in Baayen (1989, 1992) can be appropriately applied to syntactic units. Although defining exactly what units count as distinct types or as tokens of the same type in syntactic units is not as straightforward as in morphology, Zeldes (2012: 106) comes to the conclusion that a distinct type for a syntactic construction should be defined as "a distinct realization which fills all open slots in a construction, where "distinct" and "construction" should be understood in the CxG [Construction Grammar] sense, as entries in the mental lexicon." As applied to the CAN NOT VERB$_{INF}$ X$_0$ OBJECT$_{ACC}$ construction, since three open slots occur (VERB$_{INF}$, X$_0$, and OBJECT$_{ACC}$), any distinct element in any one of those slots counts as a distinct type. Thus, in English, any combination of *bear* and *tolerate* in the VERB$_{INF}$ slot, *that* and Ø in the X$_0$ slot, and *rabbit* or *rabbits* in the OBJECT$_{ACC}$ slot would count as a distinct type.

For the case at hand, the productivity of specific collexemes in the VERB$_{INF}$ can then be evaluated by "filling" the VERB$_{INF}$ slot with *stand*, *bear*, etc. to generate more specific constructions, e.g. CAN NOT STAND X$_0$ OBJECT.[13] In English, we assume for the present that the occurrence of *can not* versus *can't* is governed by phonology and speech register or other sociolinguistic considerations, not morphology or syntax per se, and thus *can not stand rabbits* and *can't stand rabbits* count as instances of the same type. Since this paper is principally concerned with the diachronic status of *can't stand* and its lexical competitors, data on type frequency and hapax legomena was only gathered for *can't stand*, *can't bear*, and *can't tolerate* (for English), and *nicht ausstehen können*, *nicht aushalten können*, *nicht leiden können*, and *nicht ertragen können* (for German).

From the same queries to the COCA corpus described in Section 5.2.1 of the main text, the number of *hapax legomena* for each of the collexemes *stand*, *bear*, and *tolerate* in the CAN NOT VERB$_{INF}$ X$_0$ OBJECT$_{ACC}$ construction was collected, which, together with the token frequencies of each colligation, allows for the calculation of Baayen's $\mathcal{P}$; these values, as well as the type frequency (V) of each colligation, are given in Table A2.

Table A2. Productivity ($\mathcal{P}$) of CAN NOT STAND/TOLERATE/BEAR.

	CAN NOT STAND X$_0$ OBJECT$_{ACC}$	CAN NOT TOLERATE X$_0$ OBJECT$_{ACC}$	CAN NOT BEAR X$_0$ OBJECT$_{ACC}$
Types (V)	779	291	348
Tokens (N)	2266	340	818
hapax legomena (h1)	512	226	229
Baayen's $\mathcal{P}$ (h1/N)	0.2259	0.6647	0.2799

The proportion of *hapax legomena* to tokens for these three collexemes in the CAN NOT construction shows that the productivity of CAN NOT TOLERATE is considerably greater than CAN NOT STAND or BEAR. This result suggests, on the one hand, that the potential of CAN NOT TOLERATE is comparatively underexploited in Present-day English, and on the other, that CAN NOT STAND or BEAR may be substantially diachronically older than CAN NOT TOLERATE. Such a conclusion is perhaps unsurprising, given that *bear* and *stand* are inherited into English from PGmc. and PIE (see Section 3 of the main text), while *tolerate* is a loanword, first attested in English in the 16th century (OED s.v. *tolerate*). The somewhat lower productivity of CAN NOT STAND as compared to CAN NOT BEAR may in turn indicate that the tendency of STAND to be colligated with the CAN NOT construction is relatively older.

Furthermore, the statistic $\mathcal{P}$ in the English data correlates strongly with the measures of colligational strength calculated in Table 8 in the main text. For instance, Pearson's *R* between $\mathcal{P}$ in Table A2 and the respective values for MINIMUM SENSITIVITY is -0.814, and between the χ^2 likelihood ratio is -0.886. In other words, the stronger the colligational strength between a given collexeme and the CAN NOT VERB$_{INF}$ X$_0$ OBJECT$_{ACC}$ construction, the lower the productivity of the constructional instantion with the collexeme. Insofar as the thesis of a relationship between smaller values of $\mathcal{P}$ and greater linguistic age is valid, the conclusion may be drawn that English *stand* is older and more established in connection with the CAN NOT VERB$_{INF}$ X$_0$ OBJECT as compared to *bear* and *tolerate*.

As far as the situation in German is concerned, Table A3 shows the same statistics given in Table A2, but for the collexemes *ausstehen*, *aushalten*, *leiden*, and *ertragen*. Due to the design of the search functions in COSMAS-II and the (in comparison to English) greater variety of syntactic structures, the types and hapax legomena for the German data had to be gathered by hand: each token of X$_0$ OBJECT$_{ACC}$ instance was input into a spreadsheet, and then filtered for duplicate values (to produce a type count) or unique values (to produce a count number of hapax legomena).

Table A3. Productivity ($\mathcal{P}$) of NICHT AUSSTEHEN/AUSHALTEN/LEIDEN/ERTRAGEN KÖNNEN

	X_0 OBJECT$_{ACC}$ NICHT AUSSTEHEN KÖNNEN	X_0 OBJECT$_{ACC}$ NICHT AUSHALTEN KÖNNEN	X_0 OBJECT$_{ACC}$ NICHT LEIDEN KÖNNEN	X_0 OBJECT$_{ACC}$ NICHT ERTRAGEN KÖNNEN
Types (V)	303	70	246	166
Tokens (N)	718	150	672	397
hapax legomena (h1)	252	63	209	137
Baayen's $\mathcal{P}$ (h1/N)	0.35097	0.42	0.31101	0.3451

Here, the resulting values for $\mathcal{P}$ seem less clearly aligned with the expectations stemming from syntactic restrictions and the measures of colligational strength reported in Table 9 in the main text, which might have predicted that $\mathcal{P}$ would be lowest for *ausstehen*. Instead, *ausstehen* is exceeded only by *aushalten* in terms of the number of distinct elements filling the object slot of the construction.

This discrepancy between the English and German results might partly be explained through further particular constructional preferences involving *aushalten*, *leiden* and *ertragen*. In the case of *aushalten*, although it had the proportionally greatest number of hapaxes, the occurrence of the pronouns *das* "that" and *es* "it" constituted much greater proportions of the tokens in comparison to *ausstehen*; with *leiden*, the same applies for *das*; with *ertragen*, the same applies for *es*, where more than one-third of all usages of *nicht ertragen können* showed *es* in the OBJECT$_{ACC}$ slot, usually as a proleptic object used to introduce a following subordinate clause headed by *wenn* "if" or *dass* "that" (e.g. *die Geschichte von der Königin, die es nicht ertragen kann, dass ihre Stieftocher Schneewittchen schöner ist als sie* "the story of the queen who can't stand it, that her stepdaughter Snow White is more beautiful than her; *Rhein-Zeitung* 20.06.2012, page 20). With *leiden*, the phrase *was ich/man nicht leiden kann* "what I/one can't stand," is exceptionally common, constituting more than 11% of all usages of *nicht leiden können* (*was* occurs as the object with *nicht ausstehen könnnen* in only about 3% of tokens). *Leiden*, too, tends occur with *es* proleptically, like *ertragen*, whereas *es* with *ausstehen* more often occurs as a "genuine" pronominal object.

The fact, however, that *aushalten* exhibits the lowest degree of colligational strength with the X_0 OBJECT$_{ACC}$ NICHT VERB$_{INF}$ KÖNNEN construction but the highest degree of productivity as measured by $\mathcal{P}$ results in a generally negative correlation between colligational strength and productivity: Pearson's *R* between $\mathcal{P}$ in Table A3 and German values in Table 9 of the main text for MINIMUM SENSITIVITY is -0.7088, and between the χ^2 likelihood ratio is -0.2777243. The German data thus confirm the same general tendency indicated in the English data, namely, that stronger collostructional strength between a lexeme and a construction correlates negatively with the productivity of the construction containing that collexeme.

The conservative conclusion to draw from this investigation of the productivity of lexically specific versions of the CAN NOT VERB$_{\text{INF}}$ X$_0$ OBJECT$_{\text{ACC}}$ construction is weak support for the hypothesis of a relationship between lower productivity and greater relative linguistic age. Somewhat surprisingly – from the point of view of this hypothesis – the differences in productivity between the German collexemes were smaller than in the English case, even though the German collexemes exhibited greater differences in degree of colligational strength and, in the case of *ausstehen*, stronger syntactic restrictions on the use of this verb (see Section 2 of the main text) are found than is the case for English *stand*. Overall, the reliability and utility of corpus-based measures of syntactic and morphological productivity as synchronic indicators of historical developments remains to be further demonstrated and qualified.

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